Case Study 17: Student Performance Trend Analysis

1. Problem Statement

Compare assessment scores across multiple examinations for individual students, calculate score differences, identify consistently declining or weak subjects, and classify students into performance categories.

2. Objectives

- Track multi-examination performance trends using lists and nested loops.
- Calculate net score improvement or decline between consecutive examinations.
- Classify student trends into categories: *Improving*, *Declining*, *Consistent High Performer*, or *Needs Intervention*.

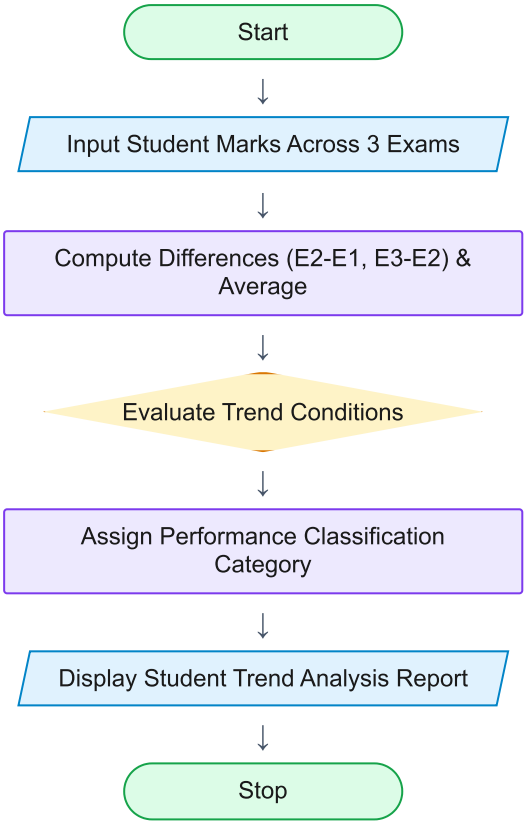
3. Assumptions

- Evaluates 3 consecutive examinations: Exam 1, Exam 2, and Exam 3.
- Subject pass mark is **40 marks** out of 100 per examination.
- A student is flagged as *Declining* if scores decrease continuously across exams.
- A student is classified as *Needs Intervention* if average score is below 50 or if any subject fails.

4. Algorithm

1. Start.
2. Read student records containing ID, Name, and list of exam marks `[E1, E2, E3]`.
3. For each student:
   - Calculate `diff1 = E2 - E1` and `diff2 = E3 - E2`.
   - Calculate net change `total_diff = E3 - E1` and average score `avg_score = (E1 + E2 + E3) / 3`.
   - If `diff1 > 0` and `diff2 > 0`, classify as **IMPROVING**.
   - Else if `diff1 < 0` and `diff2 < 0`, classify as **DECLINING**.
   - Else if `avg_score >= 80`, classify as **HIGH PERFORMER**.
   - Else if `avg_score < 50`, classify as **NEEDS INTERVENTION**.
   - Else classify as **STABLE / FLUCTUATING**.
4. Print student-wise trend evaluation report and category summary.
5. Stop.

5. Flowchart



6. Pseudocode

```
START FOR EACH student IN student_records DO E1 = student.marks[0] E2 = student.marks[1] E3 = student.marks[2] diff1 = E2 - E1 diff2 = E3 - E2 avg = (E1 + E2 + E3) / 3.0 IF diff1 > 0 AND diff2 > 0 THEN category = "IMPROVING" ELSE IF diff1 < 0 AND diff2 < 0 THEN category = "DECLINING" ELSE IF avg >= 80 THEN category = "HIGH PERFORMER" ELSE IF avg < 50 THEN category = "NEEDS INTERVENTION" ELSE category = "FLUCTUATING" ENDIF PRINT student.id, student.name, E1, E2, E3, avg, category ENDFOR STOP
```

7. Complete Python Source Code

```
      # Student Performance Trend Analysis def
analyze_student_trends(students_data): print("=" * 72) print(" STUDENT PERFORMANCE
TREND ANALYSIS REPORT") print("=" * 72) print(f"{'ID':<6}{'Name':<14}{'Exam 1':<8}
{'Exam 2':<8}{'Exam 3':<8}{'Avg':<8}{'Classification':<18}") print("-" * 72) for
student in students_data: s_id = student["id"] name = student["name"] marks =
student["marks"] e1, e2, e3 = marks[0], marks[1], marks[2] diff1 = e2 - e1 diff2 =
e3 - e2 avg_mark = sum(marks) / len(marks) if diff1 > 0 and diff2 > 0: status =
"IMPROVING" elif diff1 < 0 and diff2 < 0: status = "DECLINING" elif avg_mark >= 80:
status = "HIGH PERFORMER" elif avg_mark < 50 or min(marks) < 40: status = "NEEDS
INTERVENTION" else: status = "FLUCTUATING" print(f"{s_id:<6}{name:<14}{e1:<8}{e2:
<8}{e3:<8}{avg_mark:<8.1f}{status:<18}") print("=" * 72) # Sample Dataset across 3
Examinations records = [ {"id": "S101", "name": "Aarav Sharma", "marks": [55, 68,
82]}, {"id": "S102", "name": "Bhavna Roy", "marks": [88, 74, 62]}, {"id": "S103",
"name": "Chetan Kumar", "marks": [85, 88, 86]}, {"id": "S104", "name": "Divya
Singh", "marks": [42, 38, 45]}, {"id": "S105", "name": "Eshan Patel", "marks": [60,
75, 70]} ] analyze_student_trends(records)
```

8. Test Case and Expected Output

Evaluation Results for Sample Batch:

```
      ========================================================================
STUDENT PERFORMANCE TREND ANALYSIS REPORT
======================================================================== ID Name
Exam 1 Exam 2 Exam 3 Avg Classification ----------------------------------------
---------------------------- S101 Aarav Sharma 55 68 82 68.3 IMPROVING S102 Bhavna
Roy 88 74 62 74.7 DECLINING S103 Chetan Kumar 85 88 86 86.3 HIGH PERFORMER S104
Divya Singh 42 38 45 41.7 NEEDS INTERVENTION S105 Eshan Patel 60 75 70 68.3
FLUCTUATING
========================================================================
```

9. Conclusion

The Student Performance Trend Analysis system tracks score trajectories across multiple testing cycles, identifying students who exhibit consistent improvement, continuous decline, or require academic intervention.